

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. BACKEND & SERVER FRAMEWORK

- **Programming Language:** Python (Version 3.10+).
- **REST API Framework:** FastAPI - Chosen for its exceptional speed, built-in data validation, and native asynchronous support.
- **Web Server:** Uvicorn - The ASGI web server utilized to run the FastAPI application efficiently on the local host.
- **Security Protocol:** python-dotenv - A module used to securely manage and inject external API keys at runtime.

3. CLOUD AI INTEGRATION

- **Logical & Textual Generation:** Google Generative AI.
 - *Gemini 1.5 Flash:* Used specifically for its high-speed ability to output strict, parseable JSON arrays (the story outline).
 - *Gemini 1.5 Pro:* Used for its advanced linguistic capabilities to generate compelling dialogues and narrations.
- **Visual Artwork Generation:** Hugging Face Inference API.
 - *Stable Diffusion (v1.5):* The primary text-to-image model leveraged to render highly detailed, genre-specific comic illustrations dynamically.

4. UTILITY LIBRARIES & FRONTEND

- **Document Engine:** FPDF - A lightweight Python library used to programmatically assemble the images and text strings into a finalized PDF document.
- **Image Manipulation:** Pillow (PIL) - Used by the backend to handle image saving and to dynamically generate emergency placeholder graphics if the cloud visual API fails.

- **Frontend UI & Templating:** HTML5, CSS3, and Jinja2 - Utilized to create a clean, responsive web form and to dynamically inject backend data into the browser preview.